

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 24



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Recap of Previous Lecture



Topic

Subgroup



Topic

Cyclic group



Topics to be Covered



Topic

Properties w.r.t. Cyclic group



Topic

Some important properties in group theory





Topic : Cyclic group

* Let $(G, *)$ is a group. If there exists any element $a \in G$ such that Every element of set G can be represented in the form $(a)^n$ for some positive integer 'n', then

$$(a)^n = \underbrace{a * a * a * \dots * a}_{n \text{ times } a}$$

group $(G, *)$ is called a cyclic group. and element 'a' is one of the generator of cyclic group $(G, *)$

Note: A cyclic group may have more than one generator

Note: Let $(G, *)$ be the group and 'e' is the identity element of the group.

$$(e)^1 = e$$

$$(e)^2 = e * e = e$$

$$(e)^3 = e^2 * e = e * e = e$$

$\vdots$

Identity element of the group
can not generate any other
element except itself.

∴ Identity element can never be the generator of a group containing any other element except itself.

Q: $\{1, -1\}$ is a group w.r.t. multiplication

Here $\odot$ $1 = \text{identity}$ $\therefore$ can not generate any other element except itself

$\odot \left. \begin{array}{l} (-1)^1 = -1 \\ (-1)^2 = 1 = e \end{array} \right\}$ all the element of the group
can be generated by '-1'

Note: Order of group = 2
 $O(-1) = 2$

$\therefore \{1, -1\}$ w.r.t. multiplication is a
cyclic group and '-1' is the
generator of the cyclic group.

H.W.

① Check whether the group $\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the cyclic group

$1 = \text{identity}$ (Not a generator)
 $O(1) = 1 \neq O(G)$

$(\omega)^1 = \omega$
 $(\omega)^2 = \omega^2$
 $(\omega)^3 = \omega^3 = 1 = \text{identity}$
 $O(\omega) = 3 = O(G)$

$\therefore \omega$ is a generator

$(\omega^2)^1 = \omega^2$
 $(\omega^2)^2 = \omega$
 $(\omega^2)^3 = 1 = \text{identity}$
 $O(\omega^2) = 3 = O(G)$

$\therefore \omega^2$ is a generator

$\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group. & ω and ω^2 are generators

$$\boxed{\text{inv}(\omega) = \omega^2}$$

H.W.

②

Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the cyclic group

$$O(G) = 4$$

1 = identity
(Not a generator)
 $O(1) = 1$

$$(-1)^1 = -1$$

$$(-1)^2 = 1 = \text{identity}$$

$$(-1)^3 = -1$$

$$(-1)^4 = 1$$

$$(-1)^5 = -1$$

$$(-1)^6 = 1$$

elements will be repeated

$$O(-1) = 2 \neq O(G)$$

↳ if order of element is less than the order of the group, then it will generate the identity element before it has generated all the elements of the group, and now it will start repeating the elements that it has already generated and it can never generate the remaining elements
∴ it can not be the generator of the group

H.W.

②

Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the Cyclic group

1 = identity
(Not a generator)

$$O(1) = 1$$

$$(-1)^1 = -1$$

$$(-1)^2 = 1 = \text{identity}$$

$$(-1)^3 = -1$$

$$(-1)^4 = 1$$

$$(-1)^5 = -1$$

$$(-1)^6 = 1$$

$$O(-1) = 2 \neq O(G)$$

elements will be repeated

$$(i)^1 = i$$

$$(i)^2 = i^2 = -1$$

$$(i)^3 = i^2 \cdot i = -i$$

$$(i)^4 = i^2 \cdot i^2 = 1 = \text{identity}$$

$$O(i) = 4 = O(G)$$

all elements
 $\therefore$ generator

$$(-i)^1 = -i$$

$$(-i)^2 = i^2 = -1$$

$$(-i)^3 = +i$$

$$(-i)^4 = 1 = \text{identity}$$

$$O(-i) = 4 = O(G)$$

all elements
 $\therefore$ generator

→ Group is Cyclic,
 i & $-i$ are generators

$$\text{inv}(i) = -i$$

H.W.

③

Check whether the group $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a cyclic group or not. if cyclic then find all the generators of the cyclic group

1 - identity
(Not a generator)

$$\boxed{O(1)=1}$$

$$(3)^1 = 3$$

$$(3)^2 = 3 \otimes_8 3 = 1 \text{ identity}$$

$$O(3) = 2 \neq O(1)$$

$\therefore$ Can not generate remaining elements.

$$(5)^1 = 5$$

$$(5)^2 = 5 \otimes_8 5 = 1 \text{ (Identity)}$$

$$O(5) = 2 \neq O(1)$$

Can not generate remaining elements

$$(7)^1 = 7$$

$$(7)^2 = 7 \otimes_8 7 = 1 \text{ (identity)}$$

$$O(7) = 2 \neq O(1)$$

Can not generate remaining elements

No generator, $\therefore$ Group is not a cyclic group



Topic : Properties w.r.t. Cyclic group

① Identity element can not be the generator of the group containing any other element except itself



Topic : Properties w.r.t. Cyclic group

② Let $(G, *)$ be a group of order $= |G|$
if there exist some element $a \in G$
such that $O(a) = |G| = \text{Order of group}$
then $(G, *)$ is a cyclic group
and 'a' is one of its generator



Topic : Properties w.r.t. Cyclic group

③ Let $(G, *)$ be a group of order $= |G|$
if there exist no element $a \in G$
s.t. $O(a) = |G| = \text{order of group}$
then group is not a Cyclic group

if $O(a) \neq O(G), \forall a \in G$
then ' G ' is not a Cyclic group



Topic : Properties w.r.t. Cyclic group

④ If $a \in G$ is the generator of cyclic group $(G, *)$ then $\text{inv}(a) = a^{-1}$ will also be the generator of that cyclic group

Q:- Let $\{0, 1, 2, 3, 4\}$ w.r.t. binary opⁿ $\oplus_5$ is a cyclic group, then find all generators of the cyclic group

$O(0) = 1 \neq O(n)$
identity
 $\therefore$ Not a generator

→ group is of prime order = 5

Except for identity element, order of Every of Every element will be same as order of group.

ie. $O(1) = 5 = O(n)$

$$O(2) = 5 = O(n)$$

$$O(3) = 5 = O(n)$$

$$O(4) = 5 = O(n)$$

} all 1, 2, 3 & 4 are generators of above cyclic group

Q. Let $\{1, 2, 3, 4, 5, 6\}$ w.r.t. binary opⁿ $\otimes_7$ be a cyclic group, find all generators of the cyclic group.

$$O(G) = 6$$

1 = identity
 $\therefore$ (Not a generator)

$(2)^1 = 2$
 $(2)^2 = 4$
 $(2)^3 = (2)^2 \otimes_7 2 = 1 = \text{identity}$
 $O(2) = 3 < O(G)$
 $\therefore$ '2' is not a generator
 $\text{Inv}(2) = 4$, $\therefore$ 4 is also can not be the generator

$(3)^1 = 3$
 $(3)^2 = 3 \otimes_7 3 = 2$
 $(3)^3 = (3)^2 \otimes_7 3 = 6$
 $(3)^4 = (3)^3 \otimes_7 3 = 4$
 $(3)^5 = (3)^4 \otimes_7 3 = 5$
 $(3)^6 = (3)^5 \otimes_7 3 = 1 = \text{identity}$

* $O(3) = 6 = O(G)$
 $\therefore$ '3' is generator

* $\text{Inv}(3) = 5$
 $\therefore$ 5 is also a generator

$(6)^1 = 6$
 $(6)^2 = 6 \otimes_7 6 = 1 = \text{identity}$
 $O(6) = 2 \neq O(G)$
 $\therefore$ not a generator

* group is cyclic
and 3 & 5 are
the generators.



Topic : Properties w.r.t. Cyclic group

⑤ Let $(G, *)$ be a cyclic group of order $= n$,
and $a \in G$ is one of its generator
then

$$\{ (a)^m \mid m \in \text{Natural number and } m < n \text{ and } \text{GCD}(m, n) = 1 \}$$

is the set of all the generator of the cyclic group

$$\underbrace{a * a * a * \dots * a}_{m \text{ times } a}$$

Q:- Let $\{0, 1, 2, 3, 4\}$ w.r.t. binary opⁿ $\oplus_5$ is a cyclic group with element '1' as one of its generator then find all generators of the cyclic group

Solⁿ

$$O(G) = 5 \quad \{ \text{i.e. } n = 5 \}$$

We need to find all 'm' that are less than '5' and Co-prime to 5.

$\therefore$ 'm' values are '1', '2', '3' and 4

Hence the set of all generators will be

$$\{(a)^1, (a)^2, (a)^3, (a)^4\} \quad \text{i.e.} \quad \{(1)^1, (1)^2, (1)^3, (1)^4\}$$

i.e. $\{1, 2, 3, 4\}$ $\leftarrow$ it is the set of all generators

Q. Let $\{1, 2, 3, 4, 5, 6\}$ w.r.t. binary opⁿ $\otimes_7$ be a cyclic group. with element '3' as one of its generator

find all generators of the cyclic group

Solu.

$$O(G) = 6$$

$$\therefore n = 6$$

'm' values are '1' and '5'

* We know '3' is one of its generator

$\therefore$ Set of all generators will be $\{(3)^1, (3)^5\}$
 $= \{3, 5\}$

$$(3)^1 = 3$$

$$(3)^2 = 2$$

$$(3)^3 = 6$$

$$(3)^4 = 4$$

$$(3)^5 = 5$$



Topic : Properties w.r.t. Cyclic group

⑤ Let $(G, *)$ be a cyclic group of order $= n$,
and $a \in G$ is one of its generator
then

$$\{ (a)^m \mid m \in \text{Natural number and } m < n \text{ and } \text{GCD}(m, n) = 1 \}$$

is the set of all the generator of the cyclic group

⑥ Let $(G, *)$ be a cyclic group of order $= n$, then
number of generators of the cyclic group will be
same as number of positive integers that are less than 'n'
and Co-prime to 'n'

Euler's totient function $\phi(n)$

- Let 'n' is any fixed positive integer, then number of positive integers that are less than 'n' and coprime to 'n' can be given by Euler's totient function $\phi(n)$

$$\phi(n) = n \cdot \left\{ \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \cdot \left(1 - \frac{1}{p_3}\right) \cdots \right\}$$

$p_1, p_2, p_3, \dots$ etc are distinct prime factors of 'n'

Q: Let $(G, *)$ be a cyclic group of order = 50,
find the number of generators of cyclic group $(G, *)$

$$\text{No. of generators} = \text{No. of positive integers less than 50 and coprime to 50}$$

$$= \phi(50)$$

$$= 50 * \left\{ \left(1 - \frac{1}{5}\right) \cdot \left(1 - \frac{1}{2}\right) \right\}$$

$$= 50 * \frac{4}{5} * \frac{1}{2}$$

$$= \underline{20} \text{ Ans}$$

$$n = 50$$

$$= 25 * 2$$

$$= (5)^2 \cdot (2)^1$$

p_1

p_2

or $\{ 1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39, 41, 43, 47, 49 \}$

Q. Let $\{1, 2, 3, 4, 5, 6\}$ w.r.t. binary opⁿ $\otimes_7$ be a cyclic group. Find the number of generators of above cyclic group

$$O(n) = 6, \quad \boxed{\therefore n = 6}$$

$$\# \text{ generators} = \phi(6)$$

$$n = 6$$

$$= 3 \times 2$$

$$= (3)^1 \cdot (2)^1$$

$$p_1 \quad p_2$$

$$= 6 \times \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{1}{2}\right)$$

$$= 6 \times \frac{2}{3} \times \frac{1}{2}$$

$$= \boxed{2} \quad \underline{\text{Ans}}$$

Q. Let $(G, *)$ be a cyclic group of prime order = 'p'
find the number of generators of the group

- All the natural numbers less than 'p' are Co-prime to 'p'
and there are 'p-1' such numbers

$$\therefore \# \text{ generator} = (p-1)$$

By Euler.

$$\# \text{ generators} = \phi(p)$$

'p' is prime
 $\therefore$ only prime
factor of 'p' is
'p'. $\therefore p_1 = p$

$$= p \times \left(1 - \frac{1}{p}\right)$$
$$= \cancel{p} \cdot \frac{p-1}{\cancel{p}} = (p-1)$$



Topic : Some important properties

- ① Every group of prime order is a cyclic group.
- ② Every cyclic group is abelian group

Let $(G, *)$ be a cyclic group and 'a' is a generator

Now if $p, q \in G$ then exist some values m, n such that

$$p = (a)^m \text{ and } q = (a)^n$$

$$p * q = (a)^m * (a)^n$$

$$= \underbrace{(a * a * \dots * a)}_{m \text{ times}} * \underbrace{(a * a * \dots * a)}_{n \text{ times}}$$

because $*$ is associative

$$= \underbrace{(a * a * \dots * a)}_{n \text{ times}} * \underbrace{(a * a * \dots * a)}_{m \text{ times}} = (a)^n * (a)^m = q * p \therefore \text{Commutative}$$

- ③ Every group of prime order is abelian



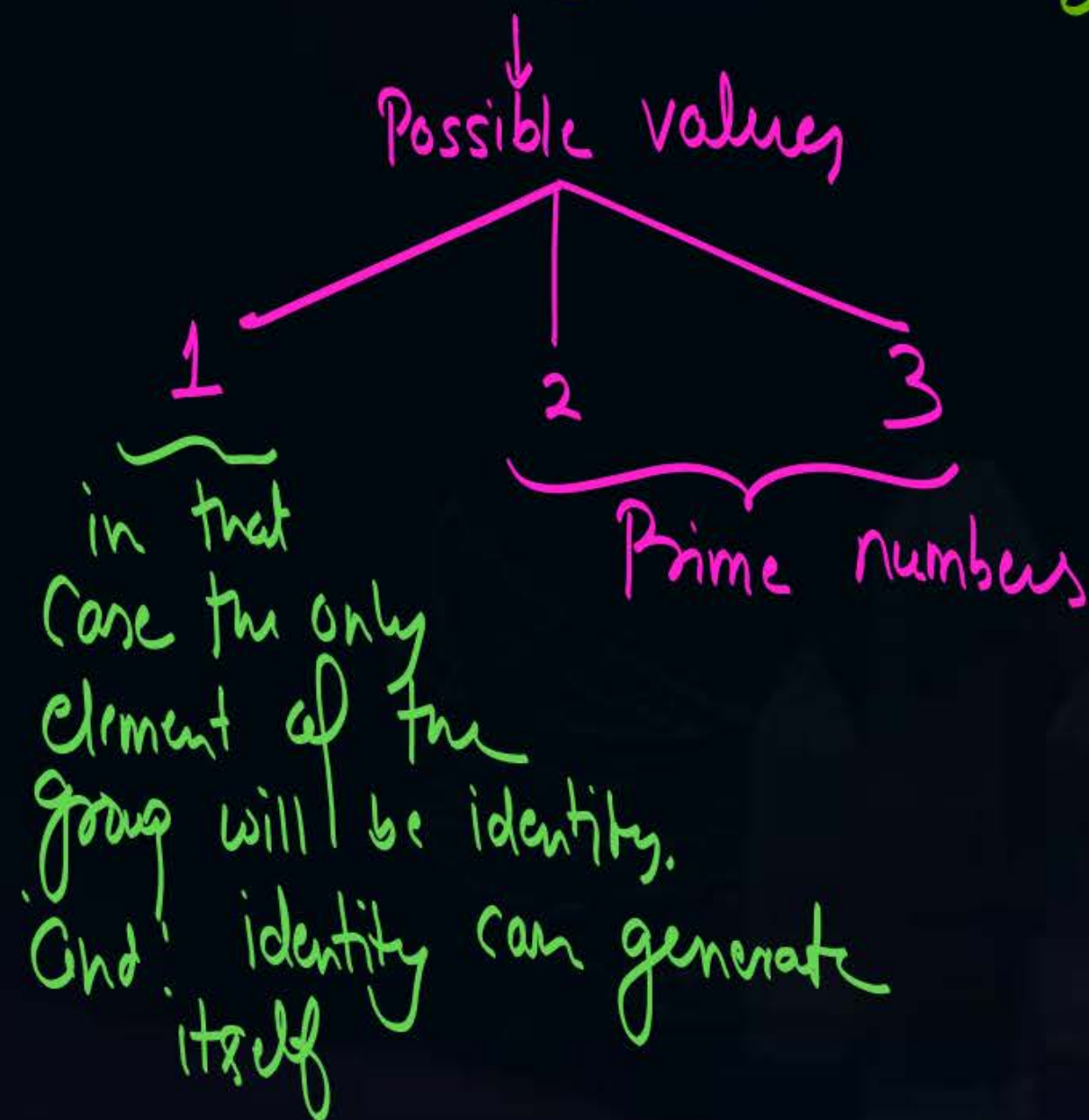
Topic : Some important properties

- ④ Every subgroup of a cyclic group is also a cyclic group, but generators of the cyclic subgroup need not be same as generator of original cyclic group



Topic : Some important properties

⑤ Every group of order ≤ 3 is always cyclic





Topic : Some important properties

⑥ Every group of order = 4 is cyclic
Except for the groups of 'Klein-4' type

Properties of "Klein-4" group:

① Order of group = 4

eg. $\{e, a, b, c\}$
identity $\swarrow$
 e

② Every element is inverse of itself

i.e.
 $\text{inv}(e) = e$
 $\text{inv}(a) = a$
 $\text{inv}(b) = b$
 $\text{inv}(c) = c$

③ Binary opⁿ on two non-identity elements results in third non-identity element

i.e.
 $a * b = b * a = c$
 $a * c = c * a = b$
 $b * c = c * b = a$

eg. $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$
is a group of Klein-4 type

identity $\swarrow$
 1
 $\{1, 3, 5, 7\}$

$\text{inv}(1) = 1$
 $\text{inv}(3) = 3$
 $\text{inv}(5) = 5$
 $\text{inv}(7) = 7$

$$3 \otimes_8 7 = 7 \otimes_8 3 = 5$$

$$3 \otimes_8 5 = 5 \otimes_8 3 = 7$$

$$5 \otimes_8 7 = 7 \otimes_8 5 = 3$$



Topic : Some important properties

- ⑥ Every group of order = 4 is cyclic
Except for the groups of 'Klein-4' type
- ⑦ 'Klein-4' group of order = 4 is not cyclic
but it is an abelian group
- ⑧ Every group of order ≤ 5 is cyclic except
for 'Klein-4' group of order = 4
- ⑨ Every group of order ≤ 5 is always abelian group



2 mins Summary



✓
Topic

Properties w.r.t. Cyclic group

✓
Topic

Some important properties in group theory

THANK - YOU